Course Outline
B.Sc. Computer Science 4- Year Programme
First Year
First Semester

Course Number	Course Title	Units
Major Courses		
COS 101	Introduction to Computer Science	3
COS 141	Computer Hardware	3
Required Ancillary Courses		
MTH 111	Elementary Mathematics I	3
MTH 121	Elementary Mathematics II	3
PHY 115	General Physics for Physical Sciences I	3
General Studies Course		
GSP 101	Study Skills and Basic Research Methods	2
GSP 111	The Use of library and Study Skills	2
Electives (Choose minimum of 3 units from the following)		
PHY191	Practical Physics I	3
CHM 101	Basic Principles of Inorganic Chemistry	2
BIO 151	General Biology I	3
STA 131	Inference I	2
Total		**22/23**

Second Semester

Course Number	Course Title	Units
Major Courses		
COS 102	Introduction to Problem Solving	3
COS 124	Introduction to Database System	3
Required Ancillary Courses		
MTH 122	Elementary Mathematics III	3
PHY 116	General Physics for Physical Sciences II	2
PHY 118	General Physics for Physical Sciences III	2
General Studies Course		
GSP 102	Basic Grammar and Varieties of Writing	2
Electives (Choose minimum of 3 units from the following)		
MTH 132	Elementary Mechanics I	3
PHY 122	Fundamentals of Physics II	3
STA 132	Inference II	2
STA 172	Statistical Computing	2
Total		**18/19**

Second Year
First Semester

Course Number	Course Title	Units
Major Courses		
COS 261	Computer Programming I	2
COS 265	Foundations of Sequential Program	2
COS 235	Compiler Construction I	2
COS 271	Data Communication and Networking	2
COS 243	Computer Architecture and Organization I	2
Required Ancillary Courses		
PHY351	Electronics	2
General Studies Courses		
GSP 201	The Social Sciences	2
GSP 207	Logic, Philosophy and Human Existence	2
Electives Courses		
(Choose minimum of 3 units from the following)		
MTH 215	Linear Algebra I	2
MTH 211	Sets, Logic and Algebra	3
STA 205	Statistics for Physical Sciences and Engineering I	2
Total		**19/20**

Second Semester

Course Number	Course Title	Units
Major Courses		
COS 236	Software Engineering I	2
COS 264	Computer Programming II	2
COS 232	Data Structures	2
COS 234	Operating System I	2
Ancillary Courses		
MTH 242	Mathematical Methods	2
General Studies Courses		
GSP 202	Issues in Peace and Conflict Studies	2
GSP 208	Nigerian Peoples and Cultures	2
Electives Courses		
(Choose a minimum of 5 units from the following)		
MTH 216	Linear Algebra II	2
MTH 222	Elementary Differential Equations I	3
PHY 242	Waves	3
STA 206	Statistics for Physical Sciences and Engineering II	2
Total		**19/20**

Third Year

First Semester

Course Number	Course Title	Units
Major Courses		
COS 361	Object Oriented Programming	3
COS 311	Switching Algebra & Discrete Structures	2
COS 331	Operating Systems II	2
COS 333	Software Engineering II	2
COS 335	Artificial Intelligence I	2
COS 341	Computer Architecture and Organization II	2
COS 337	System Analysis and Design	2
Ancillary Courses		
CED 341	Introduction to Entrepreneurship	2
Electives Courses		
(Choose a minimum of 2 units from the following :)		
COS 339	Theory of Computing	3
COS 313	Operation Research	3
COS 315	Statistical Computing	3
COS 317	Numerical Analysis	3
COS 343	Digital System Design	2
Total		**19/20**

Second Semester

Course Codes	Course Title	Units
Major Courses		
COS 382	Students Industrial Work Experience Scheme	6
COS 384	Technical SIWES Report	5
COS 386	SIWES Seminar	4
Total		**15**

Fourth Year
First Semester

Major Courses

Course Number	Course Title	Units
COS 411	Computational Science and Numerical Methods	2
COS 461	Organization of Programming Languages	2
COS 421	Database Design and Management	3
COS 431	Algorithms and Complexity Analysis	3
COS 433	Survey of Programming Languages	2
COS 441	Net-centric Computing	2
COS 463	Structured Programming	3

Electives Courses
(Choose a minimum of 2 units from the following)

Course Number	Course Title	Units
COS 435	Computer Graphics and Visualization	2
COS 413	Computer System Performance Evaluation	3
COS 415	Modeling & Simulation	3
COS 437	Project Management	2
COS 439	Game Programming	2
COS 471	Web Application Development	2
COS 423	Formal Models of Computation	3
COS 417	Optimization Techniques	3
COS 451	Laboratory for Digital System Design	3
Total		**19/20**

Second Semester
Major Courses

Course Number	Course Title	Units
COS 412	Formal Methods and Software Development	2
COS 432	Human Computer Interface	3
COS 438	Artificial Intelligence II	2
COS 490	Project	6

Required Ancillary Courses

Course Number	Course Title	Units
CED 342	Business Development & Management	2

Electives Courses
(Choose a minimum of 3 units from the following)

Course Number	Course Title	Units
COS 472	Distributed Computing System	3
COS 404	Special Topics in Computer Science	3
COS 478	Information Technology Law	2
COS 414	Queuing System Performance Evaluation	3
COS 436	Special Topics in Software Engineering	2
COS 434	Compiler Construction II	3
COS 452	Advanced Digital Laboratory	3
Total		**18/19**

Description of Courses

COS 101 Introduction to Computer Science 3 units

Concepts of hardware and software; basic subsystems of a digital computer; characteristics of a digital computer; devices/media; types/classes of computer emphasizing the PC; history of computers and their generations; Problem-solving: algorithm, flowchart, pseudo code, program/data, running a program; Programming: introduction to Java/BASIC programming language syntax and programming; Booting and shutdown; Types of operating system; User interface; managing information, computer malware (virus, etc.); computer network; Internet and web usage, e-mail; instant-messaging etc; Office applications; use of a statistical package and a database management system.

COS 102 Introduction to Problem Solving 3 units

Problem solving strategies, Concepts and properties of Algorithm; Role of algorithm in problem solving process, Characteristics of a good algorithm, steps involved in algorithm development Problem solving strategies: top down and bottom up methods; Implementation strategies, Develop Algorithm for simple problem, Representation of Algorithms; Flowchart and Pseudocodes, Development of flowcharts and pseudocodes, Flowcharts symbols and uses, Program objects. Implementation of Algorithms in any programming language such as Visual Basic, Java, C, or C++

COS 124 Introduction to Database System 3 units

Information storage and retrieval, Information management applications, Information capture and representation, analysis and indexing, search, retrieval, information privacy, integrity, security, scalability, efficiency, and effectiveness. Introduction to databases systems: components of database systems, DBMS functions, database architecture and data independence, database design and creation using a database management system. Functions of a database; uses in business and industry. introduction to the following: data dictionaries, normalization, data integrity, data modelling, and creation of simple tables, editing a database, queries, reports, and forms using SQL.

COS 141 Computer Hardware 3 units

Overview of Concepts and key terminologies; Electrostatic discharge (ESD), Cases and Form factor, Power Supply; Connectors, Voltage, Motherboard and Components, Chipsets and Form factors, CPU, Cooling System, Computer circuits, Diode array, PIAs etc, Integrated circuits fabrication process. Use of Small, Medium, Large and Very Large Scale Integrated Circuits, hardware design, Memories: Primary and Secondary memories, core memory etc, Magnetic devices; disks, tapes, video disks, etc, Peripheral devices, printers, CRTs, keyboards, character recognition. Operational amplifiers, Analog-to-Digital and Digital-to-analog converter

COS 232 Data Structures 2 units

Primitive data types; data structures: array, record, strings, list; String processing; Data representation in memory: Static and heap allocation; Runtime storage management, pointers and references; Implementation strategies for stack, queue, and hash table, graph and tree; Choosing the right data structure; High-level language data types and data-handling facilities.

COS 234 Operating System I 2 units

Overview of OS; Introduction to Operating Systems; Some examples of operating systems, Objectives of Operating Systems and Role, History of Operating Systems, Operating Systems Structure; System Components, Operating Systems Services, System Calls and System Programs, Process Management, CPU Scheduling, Scheduling Algorithms, Memory Management, Functionality Mechanisms to Support Client-server models, hand-held devices, Design Issues, Influences of security, networking, multimedia, Windows, O/S Principles: Structuring methods, Abstraction, processes of resources, Concepts of APIs, Device organization, Interrupts.

COS 243 Computer Architecture and Organization I 2 units
Fundamental building blocks, logic expression, minimization, sum of product forms, register transfer notation, physical consideration, Data representation and number bases, Fixed and floating point systems, representation memory system organization and architecture.

COS 235 Compiler Construction I 2 units
Review of compilers, assembliers, and interpreters, structure and functional aspects of a typical compiler, syntax, semantics, and pragmatics; functional relationship between lexical analysis, expression analysis and code generation. Internal form of course programme. Use of a standard compiler as a working vehicle. Error detection and recovery, Grammar and languages: the parsing problem. The scanner.

COS 236 Software Engineering I 2 units
Basic concept of software engineering, software development process: problem analysis, requirement gathering, design, implementation, testing and maintenance, choice of programming language; Object Oriented Analysis and Design (OOAD); Implementation of model/non-model based engineering processes.

COS 261 Computer Programming 2 units
Introduction to problem solving methods and algorithm development; designing, coding, debugging and documenting programs using techniques of good programming style; An IDE; The java virtual machine, Writing a java program, packages, simple Java programs; Data types; Control constructs; Classes; Keyboard Input, File I/O Using Byte Streams, Character Streams, File I/O Using Character Streams, Buffered Streams, File I/O Using a Buffered Stream, Keyboard input, file I/O using byte stream using a Buffered Stream, Writing Text Files in Java, Introduction, The File Class, Standard Streams

COS 264 Computer Programming II 2 units
Principle of good programming, structured programming concepts, structured program constructs, advantages of writing structured programs, unstructured programming construct –GOTO statement, disadvantage of writing unstructured program, data structures, debugging and testing, string processing, internal searching and sorting, recursion. Use a programming language different from that of COS201, eg. C.

COS 265 Foundations of Sequential Program 2 units
Introduction to high level language computers: meaning of high level language computers, justifications for high level language computers, arguments for and against high level language computers, high level language computer systems, high level language computer architecture, The relationship between High Level languages and the Computer Architecture that underlies their implementation; basic machine architecture, specification and translation of P/L block; Structured Languages, parameter passing mechanisms.

COS 271 Data Communications & Networking 2 units

Introduction: data and information, data processing system, data communication systems, signals; Introduction to analog and digital Time and frequency domain concepts, Fourier Transform. Introduction to Analogy and digital signals, A/D conversions, fourier series, measure of communication channel, characteristics, Nyquist and Shannon information capacity, transmission media, noise and distortion, modulation and demodulation, multiplexing and demultiplexing, synchronous and asynchronous data transmission; Error detection and control techniques. Bus structures and loop systems, computer network, Examples and design consideration, data switching principles, broadcast techniques, network structure for packet switching, protocol description of network eg ARPANET etc.

COS 311 Switching Algebra and Discrete Structures 2 units

Basic Set Theory: Basic definitions, Relations, Equivalence Relation, Partition, Ordered Sets, Boolean algebra and Lattices, Logic, Graph theory: Directed and Undirected graphs; Graph Isomorphism, Basic Graph Theorems, Matrices; Integer and Real Matrices, Boolean Matrices, Matrices med m, Path matrices, Adjacency Vectors/Matrices: Path adjacency matrix, Numerical and Boolean Adjacency matrices. Applications to counting, Discrete Probability Generating Functions.

COS 313 Operation Research 3 units

Mathematical modeling; Linear programming; Formulating a linear program, Solving linear programs; Graphical method, Simplex method; Canonical form, Two phase Simplex method , Transportation problems; Transportation Simplex Method,Network problems; Shortest Path Problem, Minimum Spanning Tree, Maximum Flow problem, Minimum-cost Flow problem, Network Simplex Algorithm; Project Scheduling Models (PSM and PERT); Game Theory; 11.1 Pure and Mixed strategies; Nonconstant-sum Games Inventory problems; Reliability problems; Dynamic programming. Network scheduling models (PERT & CPM).

COS 315 Statistical Computing 3 units

Using R as a calculator, Basics of R syntax, R workspace, Matrices and lists, subsetting. Data input and output, Interface with other software package. Writing your own codes in R, R script. Exploratory data analysis, Range, summary, mean, variance, median, sd, histogram, box plot, scatter plot. Probability distributions simulation, Conditional statements, loops and in R. Statistical functions in R: statistical inference, contingency table, chi-squared goodness of fit, regression, generalized linear models. Graphics: beyond the basics, graphics and tables, working with large datasets, principles of exploratory data analysis (Big data analysis), data frames in R.

COS 317 Numerical Analysis 3 units

Techniques for the following: Interpolation; approximation: Polynomial and splines approximation, Orthogonal polynomial and chebysev approximations. Direct and numerical integration and differentiation; solution of non-linear equations; solution of ordinary differential equations; introduction to partial differential equations. Floating-point arithmetic; use of mathematical subroutine packages; error analysis and norms; computation of eigen values and eigenvectors, related topics; numerical solution of boundary-value problems for ordinary differential equations, solution of non-linear systems of algebraic equations; least-squares solution of over-determined systems.

COS 331 Operating Systems II 2 units

Concurrency: States & state diagrams, Structures, Dispatching and Context switching; interrupts, Concurrent execution, mutual exclusion problem and some solutions. Deadlock; Models and mechanisms (Semaphores, monitors etc)Producer-consumer problem & synchronization; Multiprocessor issues; Scheduling and dispatching, ; Memory management: Overlays, swapping and partition, paging and segmentation, placement and replacement policies, working sets and trashing, caching.

COS 333 Software Engineering II 2 units
Software tools and environment: Requirement analysis and design modeling tools: systems model, process models, DFDS, state transition, state charts, Data models, ER modeling, object-oriented modeling using UML, software verification and validation, software testing; tool integration mech, software quality assurance; Programming environments, requirement analysis and design modeling tools, testing tools, tool integration mechanism. Software Design: Software architecture, design patterns, O.O. analysis & design, design for reuse. Using APIS: API programming class, browsers and related tools, Component based computing

COS 335 Artificial Intelligence I 2 units
Introduction to artificial intelligence; Problem formulation and search: heuristic search, production system; Knowledge representation: semantic network and frame, propositional logic, first order predicate logic, fuzzy logic; Other methods for reasoning; An Introduction to pattern recognition; Multilayer neural networks; Self-organizing neural networks

COS 337 System Analysis and Design 3 units
Overview of System Analysis and Design, Business System Concepts, System Characteristics and Elements, Types of Systems, Systems Models, Categories of Information, Planning and control for system success, Project Selection, Sources of project requests, Determining the user's Information Requirements and Strategies, Information gathering; Prototyping, Managing Project Review and Selection, Preliminary investigation, Conducting the Investigation, Testing Project Feasibility, System Requirement Specifications: Requirements Determination; Fact Finding Techniques; Interview, Questionnaire, Record Review and Observation, System Development Life cycle; purpose and key elements, Process Models; Waterfall, Iterative model, Agile(Scrum), Structured Analysis; Data Flow Diagram, Data Dictionary, Decision Trees, Identifying Data Requirements, Decision Tables, Pros and Cons of Each Tool, Data Modelling, Entity relationship model, Decision tree and table, Unified Modelling Language; use case diagram, sequence diagram, activity diagram etc; System design: Structure charts, Form Designs, Database design, Security, Automated tools for design.

COS 339 Theory of Computing 3 units
Chromsky hierarcy: Type 0, type 1, type 2 and type 3 grammar. Finite automata, Deterministic and non deterministic finite automata, Conversion of non deterministic finite automata to deterministic finite automata, Regular expression and their relationship to finite automata. Pushdown automata, contex free grammars: Deterministic and non deterministic pushdown automata. Context free grammars; Useless productions and emptiness set; Ambiguity; context free grammars for pushdown automata and vice-versa. Properties of context free languages, pumping lemma, closure properties, existence of non-context free languages. Turing machines, Decidability and undecidability.

COS 341 Computer Architecture and Organization II 2 units

Memory system, general characteristics of memory operation. (Technology: magnetic recording, semi-conductor memory, coupled devices, magnetic bubbles). Memory addressing, memory hierarchy, virtual memory control systems. Hardware control, micro programmed control, Asynchronous control, I/C control, Introduction to the methodology of faulty tolerant computing

COS 343 Digital System Design 2 units
Boolean Algebra: meaning, types/examples, Boolean algebra theorem/laws, using truth table to verify Boolean algebra theorem/laws; Gates, Gating devices, Combinational logic, Sequential logic, Methods of Synthesis and minimization of combinational circuit: SOP, POS, canonical forms, minterm list forms, maxterm list form, conversions from canonical forms to list term forms and vice versa, using K-map to minimize combinational circuit, Designs with multiplexer and demultiplexer, PL1, Memories latch, flip-flop, shift-register, RAM and ROM, Cache as well as microprocessors.

COS 361 Object Oriented Programming 3 units
Basic OOP concepts, Classes, Objects, Inheritance, Polymorphism, Data Abstraction, Tools for developing, compiling, interpreting and debugging Java programs, Java syntax and data objects, operators. Control flow constructs, objects and classes programming, Arrays, methods, Exceptions, Applets and the Abstract, OLE, Persistence, Windows Toolkit, Laboratory exercises in an OOP Language.

COS 382 Students Industrial Work Experience Scheme 6 units
Students spend six months on industrial attachment taken from the second semester of the third year for industrial training. A comprehensive log-book, certified by the Industrial Supervisor, and showing that the student actually participated in the SIWES

COS 384 Technical SIWES Report 5 units
At the end of the six consecutive months of Industrial training, the student is expected to submit a technical report to the Department.

COS 386 SIWES Seminar 4 units
At the end of the six consecutive months of Industrial Training, each student is expected to present a seminar on the industrial training experience emphasizing latest development in Computer Science, in relation to their industrial experience.

COS 404 Special Topics in Computer Science 3 units
Special topics from any area of computer science considered relevant at any given time. Topics are expected to change from year to year. Apart from seminars to be given by lecturers and guests, students are expected to do substantial reading on their own.

COS 411 Computational Science and Numerical Methods 2 units
Operation research: Game Theory, Network models, Inventory models. Numerical computation: Techniques for the computer implementation of the algorithms of the following: interpolation, approximation, numerical integration etc. Graphical computation, Modelling and simulation, High performance computation.

COS 412 Formal Methods and Software Development 2 units

Introduction to Formal methods in software development: Definition and meaning of formal methods, Motivations for use of formal methods in software development, Myths about formal methods in software development, Commandments of Formal Methods, The Z notation: Advantages and disadvantages, Software Formal Specification and its rationale, Software Formal specification Using Z,

COS 414 Queuing Systems Performance Evaluation 2 units
Introduction to Queuing systems: Definition of queuing systems, components of queuing systems, performance metrics of queuing systems, examples of queuing systems; Types of queuing systems: pure birth and pure death queuing systems, single server versus parallel server queuing systems, finite versus infinite queuing system, single queuing system versus queuing network; Kendall notation; Birth-death queuing models: generalized and specified models, steady state probabilities, little's law; Markovian queues up to M/G/1; Simple queues in computer networks; Bounds, inequalities and approximations.

COS 415 Modelling & Simulation 3 units
Basic definition and uses, simulation process, some basic statistical distribution theory, model and simulation, Queues; Basic components, Kendall notation, Queuing rules, Little's Law, Queuing networks, Special/types of queues, Stochastic processes; Discrete state and continuous state processes, Markov processes, Birth-Death Processes, Poisson processes, Random Numbers; types of Random Numbers, Exercises.

COS 416 Computer System Performance Evaluation 2 units
Computer system performance measures/parameters, Applications, Performance measurement techniques, simulation techniques, analytic techniques, work-load characterization, performance evaluation in selected problems, performance evaluation in design problems; evaluation of program performance.

COS 417 Optimization Techniques 3 units
Linear Programming Problem Formulations: Blending, Diet, Multiperiod, Work Scheduling, Projecvt Scheduling and Financial Optimization Problems. Solving Linear Program using Python. Simplex Algorithm: Basic and non-basic variables, multiple optimal solutions, unbounded linear programs, Degeneracy, Big-M method, Two-phase simplex method, Unrestricted variables. Sensitivity analysis. Duality Theory, Metaheuristic Algorithms.

COS 421 Database Design and Management 3 units
Relational databases: Mapping conceptual schema to relational schema; Data models; Normalization; Database connection in Linux/mysql and Windows/Access; Database access from a programming language. Practical exercises using a standard DBMS, Analysis of Open Database Connectivity standard; Stored-procedure; Object-oriented Relational Database Management Systems (e.g. Oracle) should be studied and used. Data Modelling, Relational Database Design including the use of Entity Relationship (ER) modelling using practical examples, Report generation. DBMS terminology and concepts, Transaction Management, Concurrency and Recovery; Physical design issues, Database Administrator Functions, and Distributed databases.

COS 431 Algorithms and Complexity Analysis 3 units

Basic algorithmic analysis: Asymptotic analysis of upper and average complexity bounds, standard complexity classes, time and space tradeoff in algorithms analysis, Algorithm strategies: Brute force, greedy, divide and conquer, recursive, backtracking, dynamic programming, heuristics, reduction (transform and conquer); Implementation of data structure. Fundamental computing algorithms: Numerical algorithms, sequential and binary search algorithms, sorting algorithms, binary search trees, hash tables, graphs and its representation.

COS 432 Human Computer Interface 3 units
Foundation of HCI: Definition and History of Information Architecture, Types of Information Architecture, User Centered Design, Information Architecture Models and case studies. Information Architecture Problem Solving, Information Architecture Development Process, Visual Design: Introduction to Visual Design, Design Theory, Information Design, Design Theory, Information Design, Colour, Visual Perception Theory, Principles of GUI, GUI toolkits, Human-centered software evaluation and development, GUI design and programming.

COS 433 Survey of Programming Languages 2 units
Overview of programming languages; History of programming languages, Brief survey of programming paradigms (Procedural languages, Object-oriented languages, Functional languages, Declarative-non-algorithmic languages, Scripting languages), the effects of scale on programming methodology; Language Description, Syntactic Structure (Expression notations, abstract syntax tree, Lexical Syntax, Grammar for Expression, Variants of Grammars), Language Semantics (Informal semantics, Overview of formal semantics; Denotational semantics; Axiomatic semantics; Operational semantics); The concept of types, Declaration models (binding, visibility, scope, and lifetime), Overview of type-checking, Garbage collection; Abstraction mechanisms, Parameterization mechanisms (reference vs value) Activation records and storage management, Type parameters and parameterized types. Modules in programming languages; object oriented language paradigm, functional and logic language paradigms.

COS 434 Compiler Construction II 3 units
Programming language syntax, semantics and pragmatics; Overview of translators; compilers, assemblers and interpreters; Phases of compilation; Structure of a compiler; Lexical analysis, syntax analysis, expression analysis, code generation, code optimization, error detection and recovery; Grammar and language; Construction of scanner and parser; Internal representation of a program; Recognizers, top-down and bottom-up; run-time storage organization, LR grammar and analyzers; construction of LR grammar and analyzer; construction of LR table, Organization of symbol tables; Translator writing systems.

COS 435 Computer Graphics and Visualization 2 units
Hardware aspects: Plotters, microfilm, plotters display, graphic tablets, light pens, other graphical input aid, Facsimile and its problem, Refresh display, refresh huggers, changing images, light pen interaction. Two and three dimensional transformation, perspective Clipping algorithms. Hidden line removal bolded surface removal, Warmock's method, shading, data reduction for graphical input. Introduction to hand writing and character recognition. Curve synthesis and fitting. Contouring. Ring structure versus doubly linked lists. Hierarchical structures, Data structure, organization for interactive graphics.

COS 436 Special Topics in Software Engineering 3 units

Special topics from any area of Software Engineering, considered relevant at any given time. Topics are expected to change from year to year. Apart from seminars to be given by lecturers and guests, students are expected to do substantial reading on their own.

COS 437 Project Management 2 units

Project and project life cycle; Project management processes; Structured project management; Work breakdown structures (WBS): developing WBS, using WBS to estimate effort and cost; Project scheduling: developing schedules, network diagrams, PERT and CPM, Gantt chart, using schedules to monitor and control projects; Resources and resource planning; Cost planning and budgeting; Change management; Project risk management; Ethical and commercial issues in project management.

COS 438 Artificial Intelligence II 2 units

Review of propositional and predicate logic, Resolution and theorem proving, Non-monotonic inference, probabilistic reasoning, Bayes' theorem, Natural language processing (information retrieval, language, translation, speech recognition). Machine learning (Definition and examples, supervised learning, learning, decision trees learning, neural networks, the nearest neighbour algorithm, learning theory, the problems of overfitting, unsupervised learning, reinforcement learning). Introduction to Robotics: (Overview, configuration, planning, sensing, programming, navigation and control). Expert systems; Introduction to PROLOG, LISP or any other AI programming language.

COS 439 Game Programming 2 units

Overview of Windows Programming, Overview of COM and DirectX, Mathematics of Rendering 3D Objects, Vectors and Matrices, Matrix Multiplication, Translation, Rotation, Scaling, View, Projection, Direct3D - Display Modes, Sprites and Fonts, Vertices and Transformation Matrices, Textures, Depth Buffers, Lighting. DirectX Audio - DirectSound Playback, DirectSound 3D Effects, Overview of DirectMusic. DirectInput - Mouse Input, Joystick Input, Keyboard Input, Introduction to OpenGL.

COS 423 Formal Models of Computation 3 units

Overview of the Theory of Computation ,Languages and Strings, Language Hierarchy, Task: Language Recognition, The Power of Encoding, A Machine-Based Hierarchy of Language Classes, A Tractability Hierarchy of Language Classes, Computation; Decision Procedures, Determinism and Nondeterminism , Functions on Languages and Programs, Finite State Machines and Regular Languages, FMS: Deterministic FSM, Regular Languages , Designing Deterministic FSM, Nondeterministic FSMs, Simulators for FSMs , Minimizing FSMs, A Canonical Form for Regular Languages, Finite Automata, Infinite Strings; a Regular Expression, Applications, Manipulatiom and Simplification of Regular Expressions, Regular Grammars; Regular Grammars and Regular Languages, Regular and Nonregular Languages , Algorithms and Decision Procedures for Regular Languages, Context-Free Languages (CFL) And Pushdown Automata; Rewrite Systems and Grammars , Context-Free Grammars (CFG) and Languages , Designing and Simplification of Context-Free Grammars, 11.4 Simplifying CFG, Proving That a Grammar is Correct , Derivations and Parse Trees, Ambiguity, Pushdown Automata; Definition of a Nondeterministic PDA, Deterministic and Nondeterministic PDAs, Equivalence of CFG and PDAs, Nondeterminism and Halting . Context-Free and Noncontext-Free Languages, Algorithms and Decision Procedures for Context-Free Languages, Context-Free Parsing; Lexical Analysis, Top-Down Parsing, Bottom-Up Parsing, Parsing Natural Languages, Turing Machines And Undecidability: Turing Machines; Definition, Notation and Examples, Computing With Turing Machines, Adding Multiple Tapes and Nondeterminism, Simulating a “Real” Computer, Alternative Turing Machine, The Church-Turing Thesis, The Unsolvability of the Halting Problem , Decidability and Undecidability

COS 441 Net-centric Computing 2 units

Distributed Computing: System models for distributed computing paradigm, Architectural models-Client/server model, peer-to-peer model, variations of the above model; Mobile & Wireless Computing: Introduction and overview, properties of wireless: PANs, WANs, network structure and architectures, including hand-off and mobility support at the physical/link level. ; Network Security: Foundational concepts in security: risk threats, vulnerability, advanced requirements of security; Human factors and security: ethics, regulatory environment, compliance and identity management, social engineering; Principles of secure design: isolation, fail/safe, open, end to end security; Client/Server Computing (using the web), Building Web Applications

COS 451 Laboratory for Digital System Design 3 units

General Instrumentation, Use of meter in DC and in AC measurements; rectifiers, electrodynamometer and wattmeter, instrument transformers, Oscilloscopes, Square wave, pulse, signal and function generators, wave analyzer; Logic probes/analyzers; Troubleshooting; Analyzing electronic circuits; Implementation of digital logic circuits.

COS 452 Advanced Digital Laboratory 3 units

Understanding and troubleshooting computer hardware circuits: motherboard, multi-core processors, monitor, printer, drive, card, power supplies etc. Modem, PC assembling and disassembling; Hardware and software installations; Input/Output ports, Hard disk drives, floppy disk drives, motherboards, memories and troubleshooting of digital circuits using standard equipment, Implementation of specific projects in the microprocessor laboratory.

COS 461 Organization of Programming Languages 2 units

Language definition structure, data types and structures, Review of basic data types, including lists and trees, control structures and data flow, run-time consideration, interpretative languages, lexical analysis and parsing; Functional programming language; Logic programming language, Object oriented programming language.

COS 463 Structured Programming 3 units

Structured programming elements; Models for representing structured programming elements, Structured design principles, abstraction, modularity, stepwise refinement, structured design techniques, teaching, Program modularization (bottom-up approach, top-down approach, nested virtual machine approach); Language features for structured programming; Practical illustration of above concepts using a RAD, e.g., Visual Basic; Graphical User Interface (GUI) concepts; Event-driven programming; Time manipulation; graphics; multimedia; device control, Students learn programming concepts using a project-centred approach.

COS 471 Web Application Development 2 units

Introduction to client-server, common gateway interfaces. Common web development problems and solution using server-side and client-side scripting techniques. Practical design and development of web pages using industry-standard scripting languages; XHTML, PHP, Perl; Creation of dynamic web pages. Web standards/technologies; Web2.0, XML, SOAP, web services, etc.

COS 472 Distributed Computing System 2 units

Characterization of distributed system: Definitions and meaning, Examples of distributed systems, Trend in distributed systems, Resource sharing im distributed system, challenges of distributed systems. Distributed system models: Physical models, Architectural models, Fundamental models. Networking and Internetworking: Types of network, Network principles, Internet protocols. Interprocess communication.

COS 478 Information Technology Law 3 units

Information Technology Act; Rules under Information Technology Act; The Rule of Cyberspace; Regulating Information Super highway, Cyber Law Policy Issues and Emerging Trends; Anonymous Blog Regulation versus Free Speech, Jurisdiction and the Internet; Online Contract; Digital Signatures – Legal implication; Cyber Crime; Data Protection; Liability of Intermediary; Copyright and Internet; Internet and Free Speech; Domain Name Dispute; Yahoo versus LICRA; Zippo Manufacturing versus zippo com; ProCD versus Zeidenberg; US versus Morris; US versus Thomas; Viacom versus Google.

COS 490 Project 6 units

Must involve the development of a non-trivial system or application, A report on the work done by the student will be submitted and defended before a panel of examiners consisting of members of the academic staff of the Department and possibly an external examiner.